

Theophilus Martus Goh

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SUMMARY

Computer Science (Real-Time Interactive Simulation) undergraduate with two years of full-time Unreal Engine 5 and live motion capture experience on published virtual productions. Builds real-time systems from scratch in C++ and OpenGL, including a custom game engine and a cellular-automata particle simulation. Seeking a 2027 internship in game engine, graphics, or XR/simulation engineering.

EDUCATION

B.S. Computer Science in Real-Time Interactive Simulation Sep 2024 – Apr 2028 (Expected)
Singapore Institute of Technology / DigiPen Institute of Technology *Singapore*

- Cumulative GPA: 4.02 / 5.00
- Relevant coursework: Real-Time Rendering, Computer Graphics, Modern C++ Design Patterns, Spatial Data Structures, Motion Dynamics, AI for Games, Developing Immersive Applications, Data Structures

Diploma in Game Development and Technology Apr 2017 – Apr 2021
Nanyang Polytechnic *Singapore*

TECHNICAL SKILLS

- **Languages:** C++, C, C#, Python, Java, JavaScript, GLSL, HTML/CSS
- **Engines & Real-Time:** Unreal Engine 5 (C++, Blueprints, Sequencer, Live Link Face, MetaHuman), Unity, custom C++ engine development
- **Graphics & Simulation:** OpenGL, GLSL shaders, real-time rendering, frustum culling, level of detail (LOD), spatial partitioning, AABB collision detection and response, cellular automata particle simulation
- **Virtual Production & XR:** Xsens, Rokoko, Perception Neuron motion capture, Live Link Face, morph-target facial animation, live multi-performer capture
- **Tools:** Git, Visual Studio, Blender, Maya, Google Apps Script, Excel VBA

PROJECTS

Custom Game Engine and Fighting Game | C++, OpenGL 2026

- Built a game engine from scratch in C++, implementing the rendering, input, and game object subsystems that the final title was produced on.
- Shipped a fighting game on that engine with UX and game design students, serving as technical point of contact for their content pipeline.

Sand Engine, Real-Time Particle Simulation | C++, OpenGL 2025

- Engineered a falling-sand simulation (inspired by Powder Toy and Noita) in which the environment is composed of individual particles with distinct material behaviours and interaction rules.
- Optimised the per-frame update of tens of thousands of particles to sustain interactive frame rates, the project's central engineering challenge.

Real-Time Graphics and Systems Coursework | C++, OpenGL, GLSL 2025 – 2026

- Wrote GLSL shader and rendering assignments across the programmable graphics pipeline.
- Implemented a networked multiplayer Asteroids game with client-server state synchronisation, and a MOBA lane simulator modelling emergent agent behaviour for an AI for Games module.
- Developed a LOD and frustum culling demonstrator with toggleable passes and a debug overhead view for visualising the culled frustum.

EXPERIENCE

Unreal Engine & Motion Capture Technician Feb 2022 – Feb 2024
So Drama! Entertainment / SAF Music & Drama Company (National Service) *Singapore*

- Built the Unreal Engine 5 character setup and Blueprint logic for four commissioned virtual avatars, taking them through to live deployment.
- Operated live motion capture for three public avatar shows, driving four simultaneous performers on Xsens suits with Live Link Face morph-target facial animation and troubleshooting on air.
- Benchmarked Xsens, Rokoko, and Perception Neuron under live show conditions and standardised the team on Xsens as the most stable production capture system.
- Produced music videos and social content featuring the avatars, covering scene assembly, lighting, Sequencer animation, and render-out in UE5.
- Built a company-wide equipment inventory and deployment tracking system in Google Apps Script, replacing manual logs with a shared tool accessible to all staff.

Production Operator Mar 2024 – Aug 2024
ST Engineering, Advanced Materials Engineering *Singapore*

- Operated automated packing machinery and performed QA inspection on 5.56mm, 7.62mm, and .50 BMG rounds.